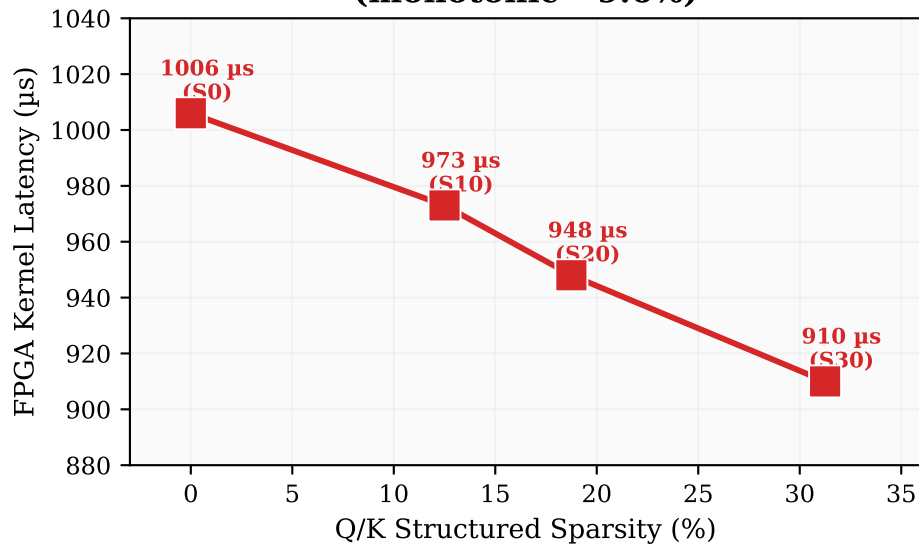
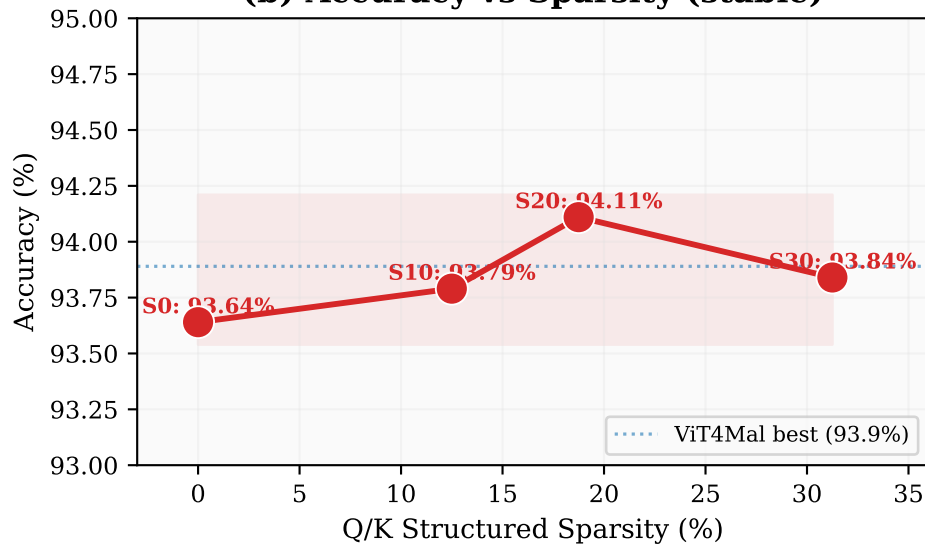


Effect of Q/K Physical Pruning on Hardware Performance

(a) FPGA Kernel Latency vs Sparsity
(monotonic -9.6%)



(b) Accuracy vs Sparsity (stable)



Physical Q/K pruning: weights W_q/W_k [ATTN_DIM $\times$ 16] (not mask). HLS loop bounds verified at csynth. PYNQ measured: kernel -9.6% , accuracy stable.